

Alireza Falah

AI RESEARCHER AND SOFTWARE DEVELOPER

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Google Scholar



“Transforming Ideas into Reality: Engineering Excellence”

Personal Profile

Ph.D. researcher specializing in developing adaptive agents using Vision Language Action Models (VLAs) in high-fidelity simulations. Proven experience in architecting high-performance C++/Python software for autonomous systems and leveraging JAX for large-scale, parallelized reinforcement learning environments. Passionate about advancing foundational AI by bridging the gap between large-scale models and complex, real-world problem-solving.

Work Experience

HUMDA Lab Nonprofit Ltd.

Budapest, Hungary

AI Researcher and Simulation Engineer

Feb 2024 - Present

- Developed and optimized **high-performance C++** and **Python** software for real-time acquisition and processing of multi-modal sensor data, including raw **LiDAR** point clouds and camera streams.
- Designed and implemented **multithreaded software components** within a complex ROS2 autonomous stack, ensuring concurrent processing of sensor data and control commands for **low-latency** system performance.
- Spearheaded the integration of **aiSim** for complex autonomous driving simulations, developing custom **ROS2 clients (C++/Python)** to manage **real-time data streaming** from virtual sensors for algorithm validation.
- Architected and maintained a complete **CI/CD pipeline** using **GitHub Actions**, **Docker**, and **Pytest** to automate testing of the C++/Python codebase, ensuring high reliability for deployment.
- Engineered advanced simulation environments in **Brax** and **MuJoCo**, leveraging **JAX** for hardware-accelerated, highly parallelized training and validation of reinforcement learning algorithms.
- Contributed to the **A2RL** competition by verifying vehicle behavior in simulators using **C++** and **ROS2**, helping the team achieve a top international ranking in a high-stakes, real-time environment.

Deep Learning Center of Iran

Tehran, Iran

AI Researcher

January 2020 - Aug 2022

- Developed and optimized high-performance computer vision perception modules (object detection, recognition, tracking) using **C++**, **Python**, and **ROS2** for real-time autonomous system applications.
- Engineered and maintained a robust **ROS2**-based software architecture for integrating advanced camera sensors and deploying computationally intensive perception algorithms on **Linux** platforms.
- Researched and implemented advanced algorithms for autonomous systems, including the creation of **simulated environments and testbeds** to rigorously test and validate perception model performance and robustness under diverse operational scenarios.
- Presented research findings and technical developments on perception systems to stakeholders and engineering teams, contributing to the organization's knowledge base in autonomous technologies.
- Technical Skills:** C++, Python, ROS2, Linux, Computer Vision, Machine Learning, Algorithm Development, Sensor Integration, Simulated Test Environments, Data Analysis.
- Soft Skills:** Team Collaboration, Innovative Thinking, Effective Communication, Problem-Solving, Analytical Rigor.

Mapna Turbine Engineering and Manufacturing Company

Karaj, Iran

Mechanical Engineer

Sep 2018 - Nov 2020

- Led the 3D design (**Autodesk Inventor**) of an Air Cooled Condenser (ACC) for S-CO₂ Power Cycles, performing detailed **Finite Element Analysis (FEA)** with **Ansys** to validate structural integrity and thermal performance.
- Developed custom analysis tools using **Matlab** to post-process **Ansys** simulation outputs.
- Utilized **Aspen Plus** for complex **process simulation** and cycle analysis.
- **Technical Skills:** Ansys (FEA), Autodesk Inventor, Aspen Plus, Matlab, Computational Analysis, Thermal & Structural Analysis, 3D Modeling, Process Simulation.
- **Soft Skills:** Teamwork, Communication, Problem-Solving, Adaptability, Project Management

Education

Eötvös Loránd University (ELTE)

Budapest, Hungary

Ph.D. in Computer Science

Sept 2024 - Present

- **Stipendium Hungaricum Scholarship** recipient.
- **Research Focus:** Developing adaptive control agents using Vision Language Action Models (VLAs) within high-fidelity physics simulation environments (e.g., **Nvidia Isaac Sim**).
- **Technical Skills:** Reinforcement Learning, Control Theory, **Nvidia Isaac Sim**, Vision Language Action Models (VLAs), Python, PyTorch.

Eötvös Loránd University (ELTE)

Szombathely, Hungary

M.S. in Automation and Industry 4.0

Sept 2022 - Jul 2024

- **Stipendium Hungaricum Scholarship** recipient.
- **Thesis:** Camera-based Tool Monitoring for Enhanced CNC Machining Processes (Supervisor: Dr. Mátyás Andó)
- Research paper based on the thesis **published** in **Discover Applied Sciences**.
- Received training from industry experts at **Schaeffler** and **TDK**. Key courses: Microcontroller Based Device Development, **Embedded Systems Programming (Assembly/Low-Level CPU-Specific)**, Industrial Communications, Digital Manufacturing, CNC Programming, Digital Factory.
- **GPA:** 4.43/5.00

Semnan University

Semnan, Iran

B.S. in Mechanical Engineering

Sept 2013 - Jul 2018

- **Thesis:** Simulation based on 'Mobile Offset Progressive Deformable Barrier: A New Approach to Cover Compatibility and Offset Testing' (Supervisor: Dr. Amir Najibi)
- Gained expertise in 3D Modeling, Solid Properties, Control and Automation, Heat Transfer, and Thermodynamics.

Publications

JOURNAL ARTICLES

Novel and Cost-Effective CNC Tool Condition Monitoring Through Image Processing Techniques

Alireza Falah, Mátyás Andó

Discover Applied Sciences (2025). 2025

CONFERENCE PROCEEDINGS

Autonomous Racing Competitions: A Review of Platforms, Software Development, and Sustainability Contributions

Zalán Demeter, Alireza Falah, Levente Puskás

Proceedings of the Conference on Sustainability (COS2024), 2024, Győr, Hungary

Achievements & Honors

Scientific Student Association (TDK in Hungarian)

Eötvös Loránd University, Hungary

Winner - University Level, Advanced to National Competition

Nov 2023

- Awarded for master's thesis at the university-level competition, advancing to the national competition.
- Thesis to be published in the university's academic journal.

TechTogether Automotive Hungary Competition

Budapest, Hungary

StudentTechLab - ELTE

May 2023 & Oct 2023

- Participated in a competitive environment where university teams solve real-world industrial problems presented by companies like **Kuka, Schaeffler, Fanuc, SMC, BPW**, and others.
- Secured 2nd place in both May and October competitions, showcasing skills in problem-solving and technical innovation.
- Led the team in an optimization task by **Fanuc**, securing 2nd place for the task and overall 4th place.
- Received a special scholarship for **consistent high performance** from university.

Projects

Exploratory Research in Large Language Models

Budapest, Hungary

Self-Directed Project

Feb 2025 - Present

- Experimented with fine-tuning and post-training of open-source LLMs (Llama series) using **Parameter-Efficient Fine-Tuning (PEFT)** techniques like **LoRA**.
- Focused on evaluating model performance on reasoning and knowledge retrieval tasks, aligning with current research in enhancing LLM capabilities.

VFD Communication and Data Collection

Szombathely, Hungary

Eötvös Loránd University

Feb 2023 - April 2023

- Developed a communication module for variable frequency drives (VFDs) using RS485 protocol.
- Implemented Python-based data collection and visualization, including a Tkinter interface for real-time monitoring and control.
- **Technical Skills:** Python, PyModbus, Modbus RTU, Tkinter.

Self-Balancing Robot

Szombathely, Hungary

Eötvös Loránd University

March 2023 - April 2023

- Designed and 3D printed a self-balancing robot using an Arduino Pro Microcontroller.
- Integrated sensors (accelerometer and gyroscope) and implemented PID control for stability.
- **Technical Skills:** 3D Printing, Arduino Programming, PID Control, Sensor Integration.

Skills

Programming	Python, C/C++, Linux, ROS2, Git, Matlab, LaTeX (Overleaf)
ML/AI Libraries	JAX, PyTorch, TensorFlow, NumPy, Hugging Face Transformers, OpenCV, Scikit-learn, Pandas
Tools & Platforms	Docker, GitHub Actions, CI/CD, MuJoCo, Brax, Nvidia Isaac Sim, aiSim, Azure
Soft Skills	Adaptability, Communication, Creativity, Critical Thinking, Presentation Skills

References available upon request.